

ForgeCloud Database Design

PostgreSQL schema, relationships, lifecycle data and audit model

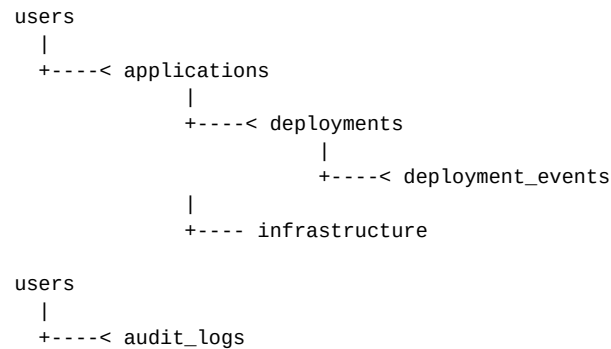
1. Database Responsibility

PostgreSQL stores users, application metadata, deployment records, infrastructure metadata, deployment events and audit records.

2. Core Tables

Table	Important fields	Purpose
users	id, name, email, password_hash, role, created_at	Identity and RBAC
applications	id, name, repository_url, branch, port, runtime, created_by	Application metadata
deployments	id, application_id, version, image, status, environment, timestamp	Release history
infrastructure	id, application_id, terraform_workspace, region, cluster_name, status	Provisioning metadata
deployment_events	id, deployment_id, event_type, message, timestamp	Deployment event stream
audit_logs	id, user_id, action, resource, resource_id, timestamp	Security/operation audit

3. Relationships



4. Data Rules

- Use primary keys for every entity.
- Use foreign keys for application/deployment relationships.
- Use timestamps for lifecycle events.
- Index common lookup fields such as email and foreign keys.
- Use constraints to prevent invalid roles and statuses.
- Use migrations rather than manual schema edits.
- Store references to secrets/configuration, not secret values where avoidable.

5. Suggested Status Values

Deployment:
PENDING
BUILDING

DEPLOYING
RUNNING
FAILED
ROLLED_BACK

6. ER Diagram Source

```
erDiagram
    USERS ||--o{ APPLICATIONS : creates
    APPLICATIONS ||--o{ DEPLOYMENTS : has
    DEPLOYMENTS ||--o{ DEPLOYMENT_EVENTS : produces
    APPLICATIONS ||--|| INFRASTRUCTURE : uses
    USERS ||--o{ AUDIT_LOGS : creates
```